

High-Speed Squeezed-State CV-QKD Over 10 km Fiber With a CMOS-Compatible Electronic-Photonic Integrated Receiver

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Abstract The bandwidth limitations of squeezed-state CV-QKD are addressed by implementing a photonic–electronic integrated receiver with a CMOS transimpedance amplifier and a broadband single-pass squeezer. A 8.45 Mbit/s asymptotic key rate is achieved over 10 km with 500 MBaud signaling rate, demonstrating scalable high-speed squeezed-state CV-QKD. ©2026 The Author(s)

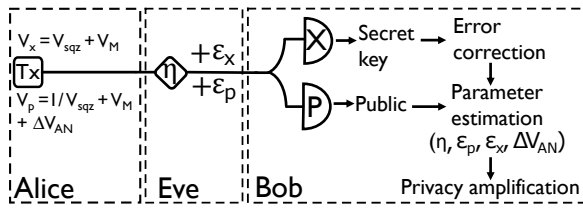


Fig. 1: Squeezed state protocol: Alice prepares squeezed vacuum states with quadrature variances V_x and V_p , where V_{sqz} is the squeezing variance, V_M is the modulation variance, and ΔV_{AN} accounts for the preparation noise. The signal propagates through Eve's channel, with loss η and quadrature-dependent excess noise $\epsilon_{x(p)}$. The p-quadrature measurement outcomes are public and used for parameter estimation, while the x-quadrature outcomes are used for key generation [1].

Introduction

Continuous-variable quantum key distribution (CV-QKD) has emerged as a promising approach for scalable quantum-secure communications, owing to its potential for high secret-key rates and intrinsic compatibility with standard telecommunication hardware [2]. Recent demonstrations have shown that CV-QKD can achieve substantially higher symbol rates than discrete-variable approaches [3, 4]. While coherent-state CV-QKD remains the dominant platform due to its experimental simplicity, both theoretical and experimental studies indicate that squeezed-state protocols provide enhanced noise tolerance and significantly relax the requirements on reconciliation efficiency [1, 5], which is a key bottleneck for real-time implementation of CV-QKD systems. In contrast to coherent-state protocols, which employ symmetric quadrature variance, squeezed-state CV-QKD exploits reduced noise in one quadrature to establish stronger correlations between the sender (Alice) and the receiver (Bob). The principle of the squeezed-state protocol used in this work is summarized in Fig. 1. As further illustrated in Fig. 2, which compares the tolerable excess-noise thresholds of the two approaches for different reconciliation efficiencies, squeezed-state CV-QKD consistently supports

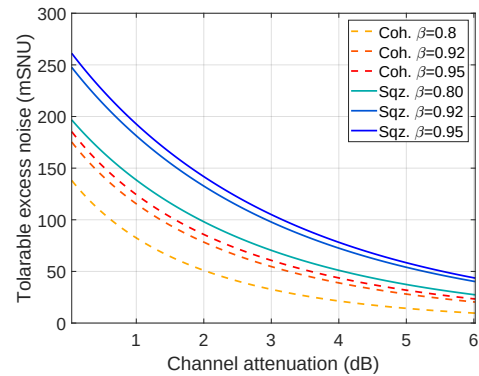


Fig. 2: Tolerable excess noise as a function of channel attenuation for coherent-state and squeezed-state CV-QKD with different reconciliation efficiency β . The modulation variance is optimized at each data point.

significantly higher noise levels across practical channel attenuations. This enhanced robustness makes squeezed-state schemes particularly attractive in realistic scenarios where environmental noise, limited reconciliation efficiency, or hardware imperfections would otherwise constrain secure key generation.

State-of-the-art squeezed-state CV-QKD implementations have successfully demonstrated practical operation but continue to rely on cavity-based squeezed-light sources characterized by limited bandwidth and bulky free-space architectures [1]. These limitations, together with large-footprint modulators and receivers, inhibit the scalability and integration potential of squeezed-state CV-QKD. Unlike cavity-based architectures, single-pass squeezed-light sources are not limited by cavity resonance and instead have their squeezing bandwidth determined only by dispersion and phase-matching conditions [6–8]. This enables terahertz-scale squeezing bandwidths, making them inherently suitable for high-symbol-rate squeezed-state CV-QKD. In parallel, integrated photonics has proven highly effective for realizing compact and scalable CV-QKD systems, offering high-speed modulation and re-

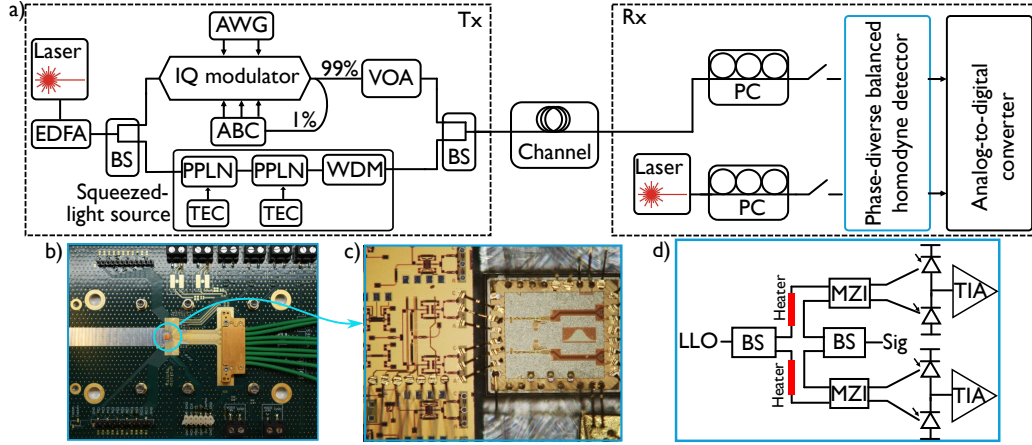


Fig. 3: (a) Experimental setup. (b) Receiver PCB. (c) Micrograph of the receiver. (d) Schematic of the phase-diverse receiver. Acronyms: erbium-doped fiber amplifier (EDFA), periodically poled lithium niobate (PPLN), wavelength-division multiplexing (WDM), arbitrary waveform generator (AWG), variable optical attenuator (VOA), automatic bias controller (ABC), beam splitter (BS), thermo-electric coolers (TEC), polarization controller (PC), Mach-Zehnder interferometer (MZI).

ceiver architectures well matched to broadband squeezed light [3, 4, 9]. Combining these technologies opens a path to practical, high-speed squeezed-state CV-QKD implemented entirely on integrated platforms.

In this paper, we present the first high-speed squeezed-state CV-QKD system enabled by a broadband single-pass squeezed-light source and a photonic–electronic integrated receiver based on integrated photodiodes and a CMOS transimpedance amplifier (TIA). Leveraging the terahertz squeezing bandwidth of the single-pass source and the high-speed capabilities of the receiver front-end, the proposed system operates at 500 MBaud signaling rate, i.e., an order of magnitude faster than state-of-the-art cavity-based implementations [1]. This approach achieves an asymptotic secret key rate of 8.45 Mbit/s over a 10 km fiber channel.

Squeezed-state CV-QKD protocol

Alice and Bob used the squeezed quadrature to generate secret keys while the anti-squeezed quadrature measurements are publicly announced. The implemented protocol is outlined in references [1, 10]. The asymptotic secret key rate K_∞ against collective attack reads:

$$\begin{aligned} K_\infty &= R(\beta I_{AB} - \chi_{BE}) \\ &= R(\beta I_{AB} - [S(A, B_x|B_p) - S(A|B_x, B_p)]) \end{aligned}$$

where R is the system repetition rate, β is the classical reconciliation efficiency, χ_{BE} is the upper bound of the information accessible to Eve, I_{AB} is the mutual information between Alice and Bob, $S(\cdot|\cdot)$ is the conditional von Neumann entropy on the quadrature measurements, A and $B_{x(p)}$ represent Alice's and Bob's quantum random variables in the corresponding quadrature. With this approach, we can do a practical intradyne detection using a locally generated local oscillator (LLO) and still have the advantages of squeezed states.

Experimental setup

The assembly with the integrated photonic-electronic receiver is shown in Fig. 3 together with the full experimental setup. The squeezed-light source consists of two periodically poled lithium niobate (PPLN) waveguides, each temperature-stabilized using dedicated thermo-electric coolers (TECs) to ensure stable phase matching. An erbium-doped fiber amplifier (EDFA) allows to boost the C-band laser optical power to the required pump levels for efficient squeezed light generation. The first PPLN stage, pumped at 1550 nm, generates 775 nm light via second-harmonic generation (SHG). This up-converted SHG light subsequently pumps the second PPLN stage, operated in the reverse direction, to generate squeezed light. A wavelength-division multiplexing (WDM) coupler filters out the residual 775 nm pump light at the output of the squeezed-state transmitter.

The quantum signal was generated by a quantum random number generator (QRNG), and up-sampled using a root-raised-cosine filter. The system operates at a repetition rate of 500 MBaud. Two frequency-multiplexed pilot tones are added for digital signal processing (DSP). Due to the low-frequency cut-off of the receiver, double-sideband modulation is used, with the quantum signal up-shifted to 1.5 GHz to avoid spurious electromagnetic interference (EMI) contributions in the received signal spectrum. Waveforms are generated with an arbitrary waveform generator (AWG) operating at 5 GSa/s. The transmitter consists of an IQ modulator followed by a variable optical attenuator (VOA). The IQ modulator is controlled using an automatic bias controller (ABC). The coherent states generated by the transmitter chip were mixed with the squeezed vacuum states from the squeezed light source on a 90:10 beam splitter (BS) to generate displaced squeezed states.

Tab. 1: Estimated system parameters from transmission experiments. Asymptotic SKR calculated assuming 95% reconciliation efficiency. V_{sqz} : squeezing variance, V_{AN} : anti-squeezed noise variance, V_{mod} : modulation variance, $\varepsilon_{x,p}$: excess noise referred to channel input, η : channel efficiency, V_D : trusted noise referred to channel output, where SNU denotes the shot-noise unit.

Fiber (km)	V_{sqz} (SNU)	V_{AN} (SNU)	V_{mod} (SNU)	ε_x (SNU)	ε_p (SNU)	η	V_D (SNU)	SKR (Mb/s)
10	0.679	0.726	0.700	0.014	0.108	0.415	0.078	8.45

The quantum channel consists of a 10 km fiber spool, assumed to be fully accessible by an adversary. At the receiver, an independent laser provides the LLO. Because the squeezed-light source supports terahertz-scale optical bandwidths, RF-heterodyne detection is impractical. Instead, intradyne detection was employed using a phase-diverse receiver, enabling simultaneous detection of both conjugate quadratures.

The receiver is composed by a photonic integrated circuit (PIC) fabricated using imec's silicon photonic (SiPh) platform. It consists of a phase-diverse balanced homodyne detector (BHD) with integrated silicon-germanium (SiGe) photodiodes characterized by 88% quantum efficiency. The two balanced photodiode (PD) pairs are directly interfaced with a two-channel TIA implemented in a 28-nm CMOS platform. The two ICs are co-packaged on a custom high-speed printed circuit board (PCB) with coaxial connectors for the high-speed output signals readout.

Light was coupled into the signal and the LLO ports of the BHD through individual grating couplers (GCs), each exhibiting approximately 2.5 dB insertion loss. Polarization controllers (PCs) were used to inject the optimal polarization in each GC. The receiver achieves shot-noise-limited operation with a bandwidth exceeding 10 GHz and a common-mode rejection ratio (CMRR) above 30 dB across the whole usable bandwidth. The total efficiency of the receiver is around 0.46. Two optical switches were used for the calibration of the vacuum and the electronics noise. The detected signals are sampled by an analogue-to-digital converter (ADC) at 5 GSa/s. A subsequent DSP chain synchronizes the digital clocks and recovers key parameters, including the carrier frequency and phase, the squeezing angle, and the relative phase between the squeezed light and the coherent state [1].

Results

The estimated system parameters are summarized in Table 1. In total, 10^7 quantum symbols were exchanged between Alice and Bob. The quantum signal is centered at 1.5 GHz, while the optical beating between the LLO and the transmitter laser appears at approximately 50 MHz. Two pilots were used to recover the clock offset between the transmitter and receiver digital systems [11, 12]. The relatively low channel efficiency is primarily attributed to mismatches in polarization alignment and a large loss in the fiber spool. The

excess noise in the p -quadrature might also arise from fluctuations in the optical probing which can induce variations in the detector noise floor.

Fig. 4 shows the asymptotic key rate after the 10 km channel. Assuming a reconciliation efficiency of 95%, our system achieves a secret key rate of 8.45 Mbits/s in the asymptotic regime [10].

Conclusion

Squeezed-state CV-QKD offers fundamental performance advantages over coherent-state protocols, yet research activity has remained limited due to the complexity of bulk-optic implementations and the restricted operating speeds imposed by cavity-based squeezing sources. In this work, we demonstrate a CV-QKD system that combines a fiber-based broadband squeezed-light source with a bulk IQ-transmitter and an photonic-electronic integrated phase-diverse receiver. Operating over a 10 km fiber link, our system achieves an asymptotic key rate of 8.45 Mbit/s. These results demonstrate a practical route toward scalable, high-rate CV-QKD architectures that leverage age squeezed light to enhance performance.

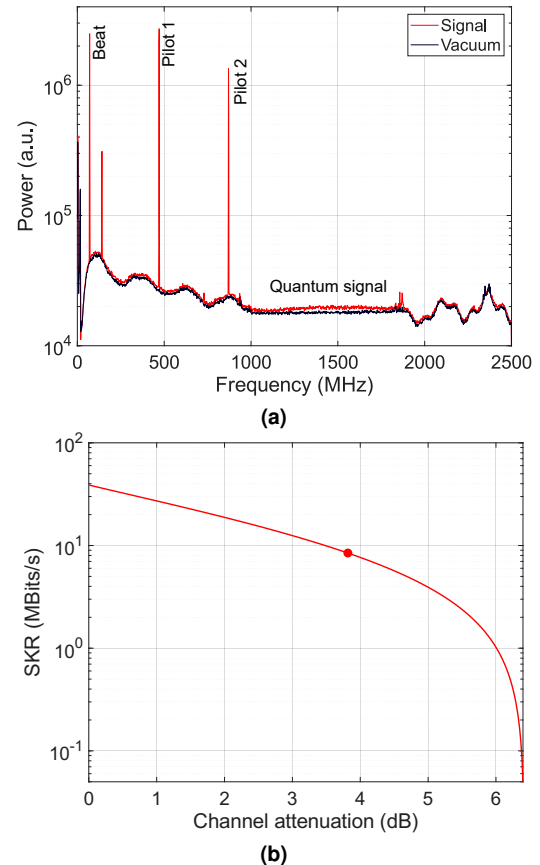


Fig. 4: (a) Power spectrum of the received signal in one of the receiver channels after applying the whitening filter. (b) Secret key rate after 10 km fiber channel.

Acknowledgements

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